Proposal: Evidence-Grounded Multimodal Models for Scientific Execution

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Research Problem

Scientific and clinical work is not just a written protocol or a polished final result. It unfolds through video, instruments, timestamps, operator choices, deviations, repeated attempts, and uncertainty. I propose to build a multimodal scientific-execution model that aligns procedure video, protocol text, event traces, and outcome labels into an evidence-grounded timeline of what happened, what may have deviated, and which observations support each conclusion.

Why It Matters

My current research in ophthalmic surgical-video analysis has made this gap concrete: phase recognition, tool detection, event detection, retrieval-grounded report generation, and question answering are useful only when a model can expose its evidence and uncertainty. Transfyr's domain generalizes this challenge to laboratory science, where hidden execution details can determine whether an experiment is reproducible, safe, and interpretable.

Approach

- Construct a process-level representation that links video segments, protocol steps, instrument/time signals, operator actions, and result metadata into a long-horizon execution memory.
- Train retrieval-grounded models that answer what happened and why, returning supporting snippets or time intervals rather than ungrounded summaries.
- Model conflicting evidence explicitly through calibrated uncertainty, contradiction-aware retrieval, and attribution over sensors, people, and protocol context.
- Design evaluations where success means fewer hidden execution errors, stronger expert review, and better transfer across sites or procedures, not just benchmark accuracy.

Fit With Transfyr

This project matches Transfyr's agenda in multimodal reasoning, conflicting evidence, long-context scientific memory, real-world evaluations, and tacit expertise. I would bring experience with biomedical computer vision, surgical-video datasets, retrieval-grounded reporting, multi-agent reasoning, and optimization. The fellowship would let me move from clinical-video AI toward a broader observability layer for real scientific execution.

Resources Needed

I would need access to Transfyr's multimodal scientific-execution data, expert feedback from scientists/operators, and GPU compute for video-language modeling and retrieval experiments. A reasonable starting budget would be 4-8 modern GPUs for iterative prototyping, plus storage for video and trace data. The final artifact would be a publishable method, evaluation benchmark, and demo for evidence-grounded scientific-execution review.